Coordinate Geometry & Transformations Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Distance formula

ORIGINAL PROBLEM

Find the distance between (1, 2) and (4, 6).

INCORRECT WORKED SOLUTION

$d = |4 - 1| + |6 - 2|$

$d = 3 + 4 = 7$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Midpoint formula

ORIGINAL PROBLEM

Find the midpoint of (−2, 6) and (4, −2).

INCORRECT WORKED SOLUTION

$M = \left(\frac{4 - (-2)}{2}, \frac{6 - (-2)}{2}\right)$

$M = (3, 4)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Classifying quadrilaterals

ORIGINAL PROBLEM

Is ABCD with A(0, 0), B(4, 0), C(3, 3), D(1, 3) a parallelogram?

INCORRECT WORKED SOLUTION

Slope of AB = 0 and slope of DC = 0.

AB ∥ DC.

So ABCD is a parallelogram.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Reflection

ORIGINAL PROBLEM

Reflect (3, −2) across the x-axis.

INCORRECT WORKED SOLUTION

Rule: (x, y) → (−x, y)

(3, −2) → (−3, −2)

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Reflection over y = x

ORIGINAL PROBLEM

Reflect (2, 5) across the line y = x.

INCORRECT WORKED SOLUTION

Rule: (x, y) → (−y, −x)

(2, 5) → (−5, −2)

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Rotation

ORIGINAL PROBLEM

Rotate (4, 1) 90° counterclockwise about the origin.

INCORRECT WORKED SOLUTION

Rule: (x, y) → (y, −x)

(4, 1) → (1, −4)

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Translation

ORIGINAL PROBLEM

Translate $(-3, 5)$ by the vector $\langle 4, -7 \rangle$.

INCORRECT WORKED SOLUTION

$(-3 - 4, 5 + 7)$

$(-7, 12)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Dilation

ORIGINAL PROBLEM

Dilate (2, 1) by a scale factor of 3 about the origin.

INCORRECT WORKED SOLUTION

$(2 + 3, 1 + 3)$

$(5, 4)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Dilation (shrinking)

ORIGINAL PROBLEM

Dilate $(6, -4)$ by a scale factor of $\frac{1}{2}$ about the origin.

INCORRECT WORKED SOLUTION

$(x, y) \rightarrow (2x, 2y)$

$(6, -4) \rightarrow (12, -8)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Equation of a circle

ORIGINAL PROBLEM

Write the equation of a circle with center $(-2, 3)$ and radius 4.

INCORRECT WORKED SOLUTION

$(x - h)^2 + (y - k)^2 = r^2$

$(x - 2)^2 + (y + 3)^2 = 16$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)